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Date: 12 Nov 2026

21LEM202T — Unit 4, Session 8, SLO 1

Activity B1 — Fill in the water figures from the slides.

Statement	Your answer
India's total annual rainfall ≈ **4,000** cubic km.	**4,000**
Total annual human water use ≈ **600** cubic km.	**600**
Underground aquifer storage ≈ **400** cubic km.	**400**
In 2014 rainfall was only **88** % of normal.	**88**
Of 5,723 groundwater blocks, **1,734** (**30** %) are semi-critical/critical/over-exploited.	**1,734 (30)**
60 % of aquifers may be critical by 2028 if the trend continues.	**60**

Activity B2 — Natural vs Unnatural storage/distribution of water.

Classify each item as Natural or Unnatural.

Items: Forests • Dams • Underground aquifers • Canals • Rivers & streams • Glaciers

Natural storage / distribution	Unnatural storage / distribution
Forests	**Dams**
Underground aquifers	**Canals**
Rivers & streams	
Glaciers	

Activity B3 — Fill in the food-and-resource figures.

Statement	Your answer
1 kg of meat needs **3** kg grain and **15,000** litres of water.	**3 kg grain, 15,000 litres**
The world's cattle alone consume grain enough for **8** billion people.	**8**
Land use: **2** acres/person (meat) vs **0.5** acres/person (veg).	**2 vs 0.5**
Of food produced (4.2 billion tons), about **1.3 billion tons** is wasted.	**1.3 billion tons**
Global food production is about **1.5** times the requirement.	**1.5**

PART C Long-Answer Questions

Q1. India has abundant natural fresh water yet faces water insecurity. Using the "abundance vs mismanagement" comparison, explain the main human causes of this insecurity.

India receives ample rainfall and has extensive river systems, but water insecurity arises from mismanagement: over-extraction of groundwater, pollution of surface water bodies, inefficient irrigation practices, lack of rainwater harvesting, and poor maintenance of traditional water storage structures. These human actions create scarcity despite natural abundance.

Q2. “Food insecurity is not a question of production.” Justify this using the FAO wastage figures and the idea of right utilisation / right education-sanskar.

Global food production is about 1.5 times the requirement, yet 1.3 billion tons of food is wasted annually. This shows that the problem is not insufficient production but improper distribution, wastage, and lack of right utilisation. Right education and sanskar (values) would teach people to avoid waste, share surplus, and consume only what is needed, thereby eliminating food insecurity.

Q3. Explain the statement “The problem is not just the population; the problem is population without right understanding,” using the three factors listed for population growth.

The three factors for population growth are: (1) lack of education about family planning, (2) poverty leading to children as economic assets, and (3) cultural/religious norms favouring large families. Without right understanding — i.e., awareness of resource limits, responsible parenthood, and sustainable living — population growth strains resources. With right understanding, even a large population can live harmoniously within ecological limits.

Self-Reflection (Take-Home) / Estimate your household’s daily water use and note two ways to reduce indiscriminate use. / For one day, record the food you waste and reflect on “right utilisation.”

My household uses about 500 litres of water daily. Two ways to reduce indiscriminate use: (1) fix leaking taps and use low-flow showerheads, (2) collect rainwater for gardening. For one day, I wasted about 200g of cooked rice and half a chapati. Reflecting on right utilisation, I realise I should take only what I can eat and store leftovers properly, avoiding waste and respecting the resources used to produce the food.